

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int pages[30], frame[10], time[10];
```

```
    int n, f, i, j, k, pos, fault = 0, hit;
```

```
    printf("Enter number of pages: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter reference string:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &pages[i]);
```

```
    printf("Enter number of frames: ");
```

```
    scanf("%d", &f);
```

```
    for(i = 0; i < f; i++)
```

```
        frame[i] = -1;
```

```
    printf("\nPage\t");
```

```
    for(i = 0; i < f; i++)
```

```
        printf("F%d\t", i + 1);
```

```
    printf("Status\n");
```

```
    for(i = 0; i < n; i++)
```

```
{  
    hit = 0;  
  
    for(j = 0; j < f; j++)  
    {  
        if(frame[j] == pages[i])  
        {  
            hit = 1;  
            time[j] = i;  
            break;  
        }  
    }  
}
```

```
if(!hit)  
{  
    pos = -1;  
  
    for(j = 0; j < f; j++)  
    {  
        if(frame[j] == -1)  
        {  
            pos = j;  
            break;  
        }  
    }  
}
```

```
if(pos == -1)
{
    pos = 0;

    for(j = 1; j < f; j++)
    {
        if(time[j] < time[pos])
            pos = j;
    }
}
```

```
frame[pos] = pages[i];
time[pos] = i;
fault++;
}
```

```
printf("%d\t", pages[i]);
```

```
for(k = 0; k < f; k++)
{
    if(frame[k] == -1)
        printf("-\t");
    else
        printf("%d\t", frame[k]);
}
```

```
    if(hit)
        printf("Hit\n");
    else
        printf("Fault\n");
}

printf("\nTotal Page Faults = %d\n", fault);

return 0;
}
```